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Preparation of *Ocimum tenuiflorum* essential oil loaded bioactive nanoformulation

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ABSTRACT

Essential oil based nano-formulations are need of the time as the efficacy and stability of such formulations have valuable applications in the field of medicine and cosmetic research. Hence, in present study, the essential oil from Ocimum tenuiflorum plant was extracted by hydro-distillation method. The compositional ingredients used to formulate nanoemulsion by low energy spontaneous method. The physical characteristics of emulsion were evaluated by analyzing pH, viscosity, conductivity and surface tension. The droplet size and zeta potential of the nanoemulsion was analyzed by using a Zetasizer. The essential oil was then added in the prepared nanoemulsion in different concentrations, resulting in the formation of 5 samples (F1 to F5). The samples showed significant antioxidant activity as tested by DPPH method. Mosquitoes' repellency test of these essential oil loaded nanoemulsions showed that the activity increase with increasing e. oil concentration with 100 % repellency for F5. The results of present study suggested that the prepared formulation can be used as an important ingredient to formulate mosquito repellent products.

PROBLEM STATEMENT

The essential oils are mixtures of naturally occurring phytochemicals with broad range of biological activities and potential applications in various medicinal and cosmetic products. The challenges associated with using essential oils via traditional means include high volatility and their poor solubility in water, which can limit their effectiveness. The use of nanoemulsion technology can overcome these issues, as it can increase the stability and solubility of essential oils and while also enhancing their bioavailability and stability. Hence, this study was aimed to address the research question that "Can essential oil-based nanoemulsion be developed as an effective and sustainable bioactive agent?"

APPLICATIONS

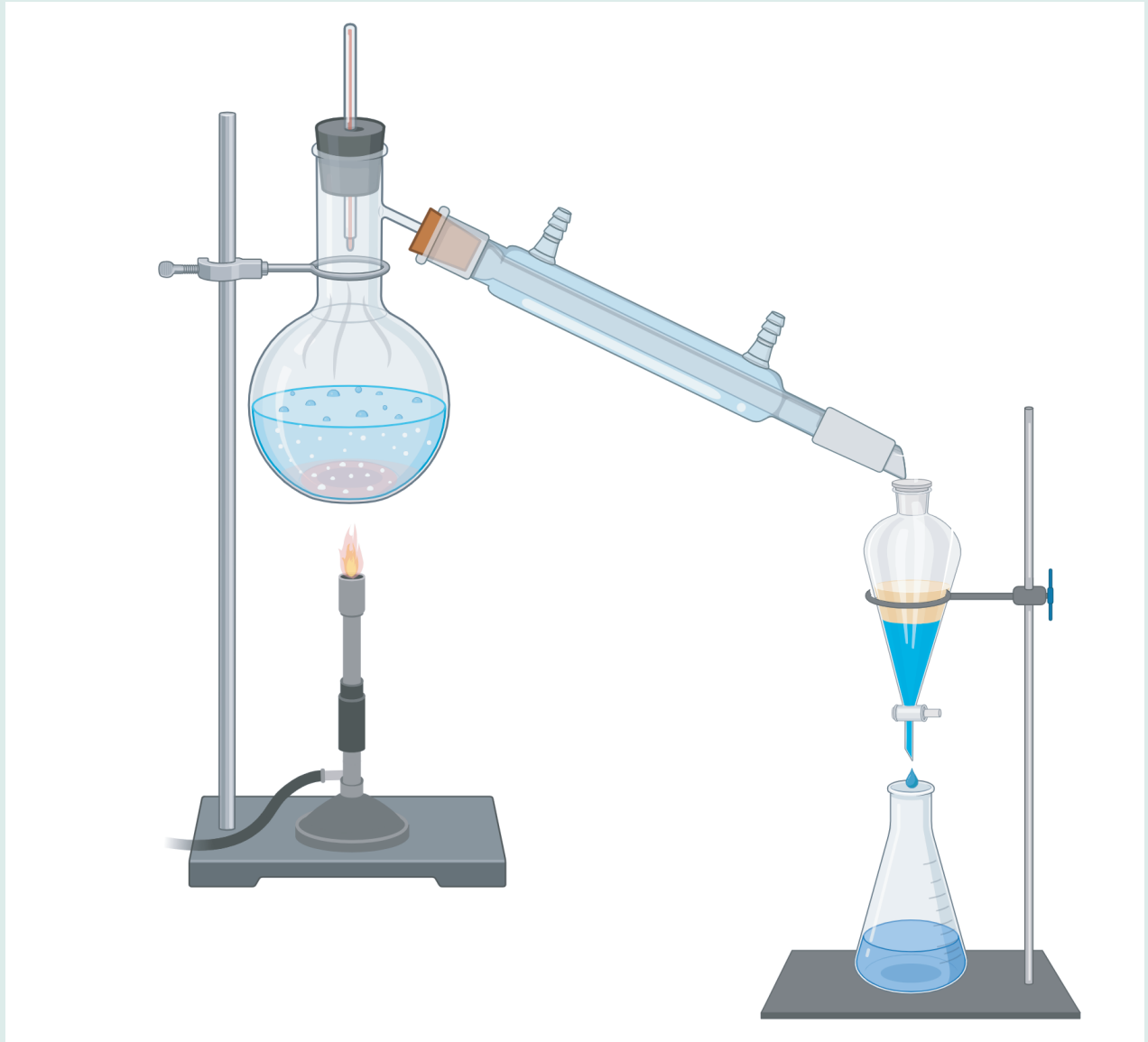
The results of our study indicated that *Ocimum tenuiflorum* essential oil loaded nanoemulsion can be used as:

- Natural Mosquito repellent product
- Natural antioxidant in cosmetic products
- Natural ingredient in anti-inflammatory products

METHODOLOGY

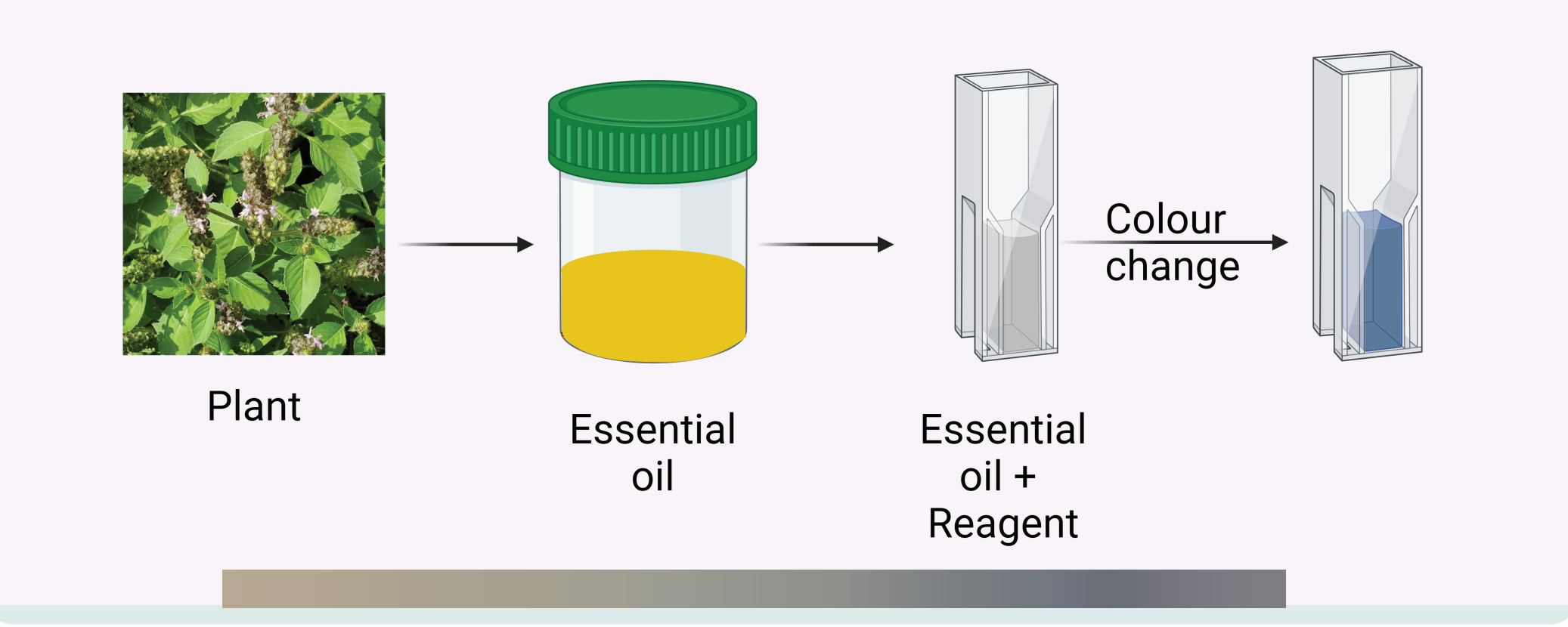
Plant

Hydrodistillation

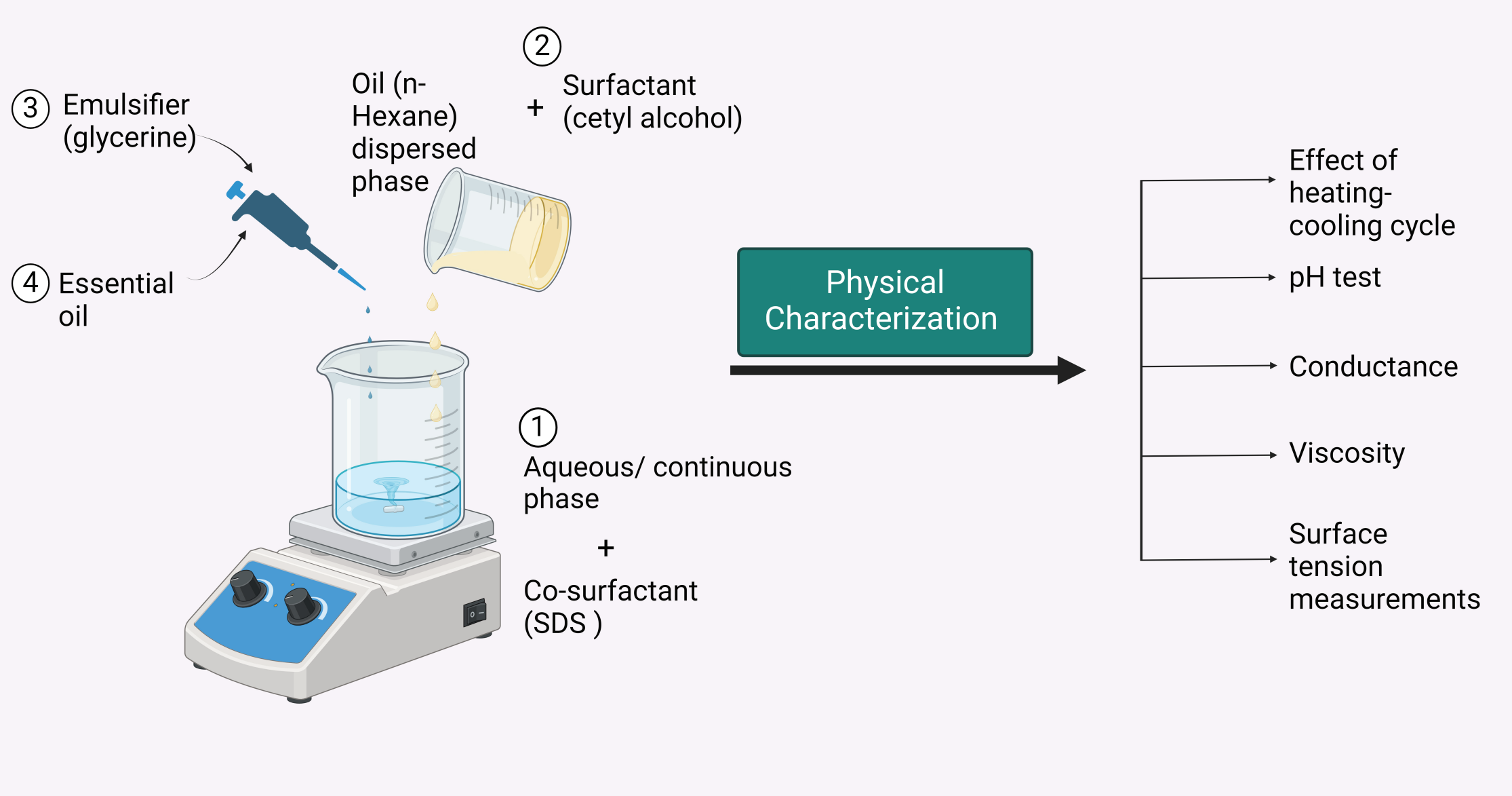


GC MS

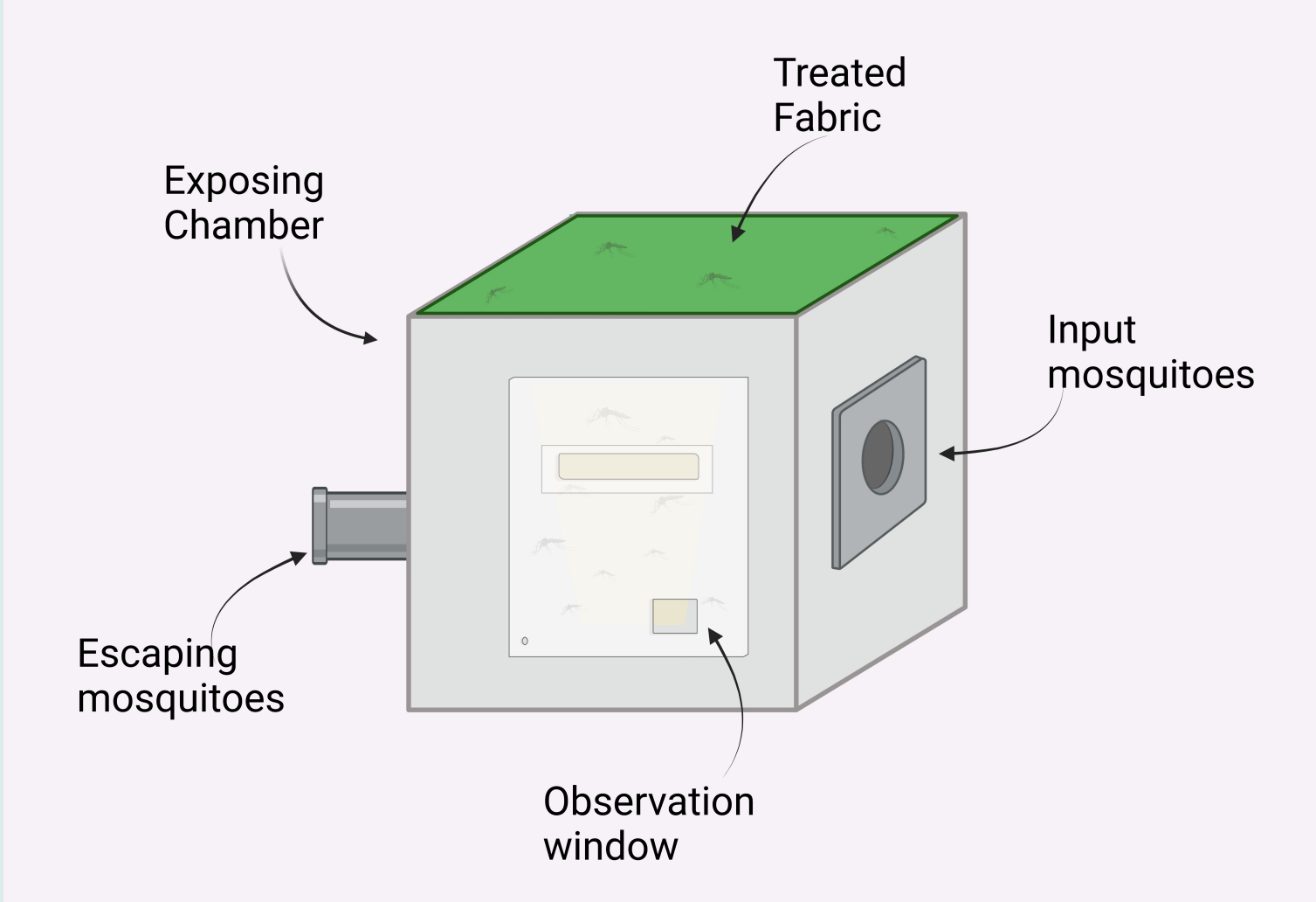
Total phenolic contents of essential oil



Formulation and analysis of physical parameters


③ Emulsifier (glycerine)
Oil (n-Hexane) dispersed phase
② + Surfactant (cetyl alcohol)
④ Essential oil
① Aqueous/ continuous phase
+
Co-surfactant (SDS)
Physical Characterization
Effect of heating-cooling cycle
pH test
Conductance
Viscosity
Surface tension measurements

Mosquitoes' repellent activity of nanoemulsion

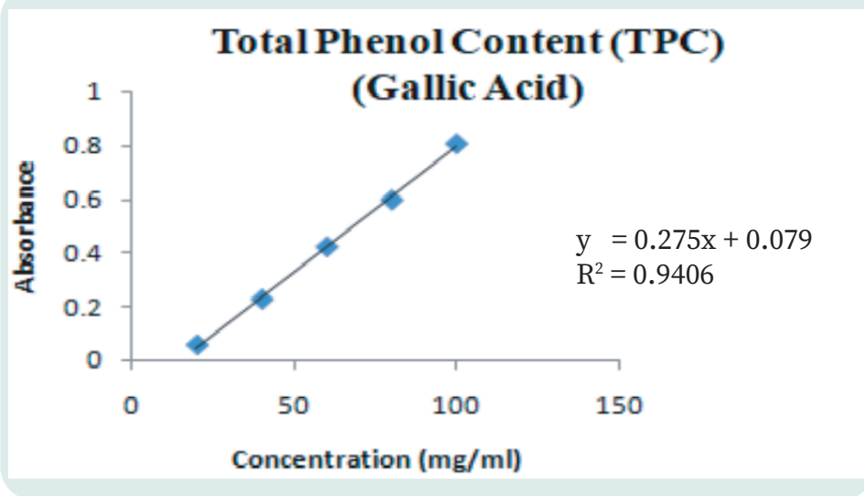


Zetasizer

RESULTS

The yield of essential oil
0.09 ± 0.01 percent

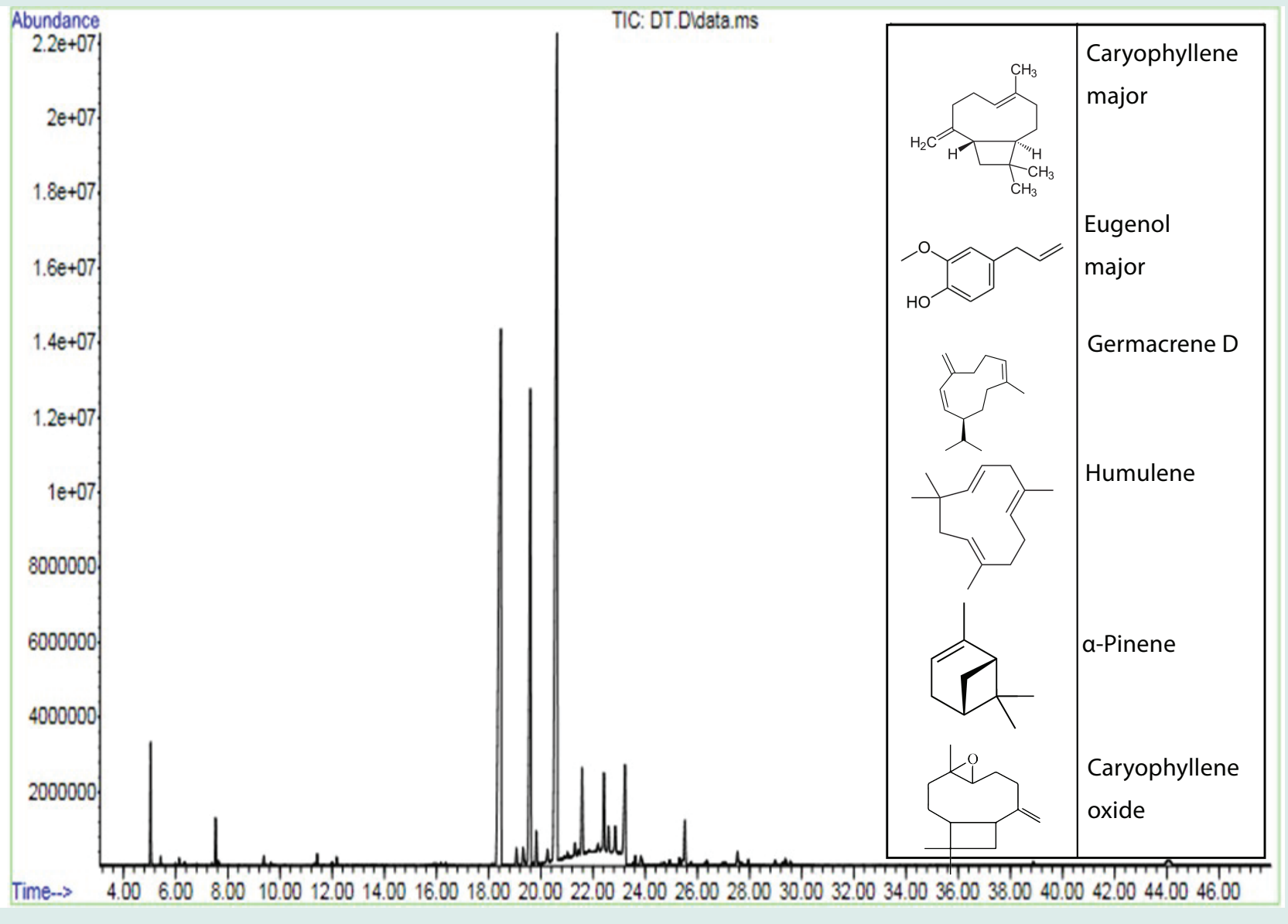
Total phenolic contents
246.3 ± 1.15 mg equivalents of gallic acid per gram dry weight of the plant



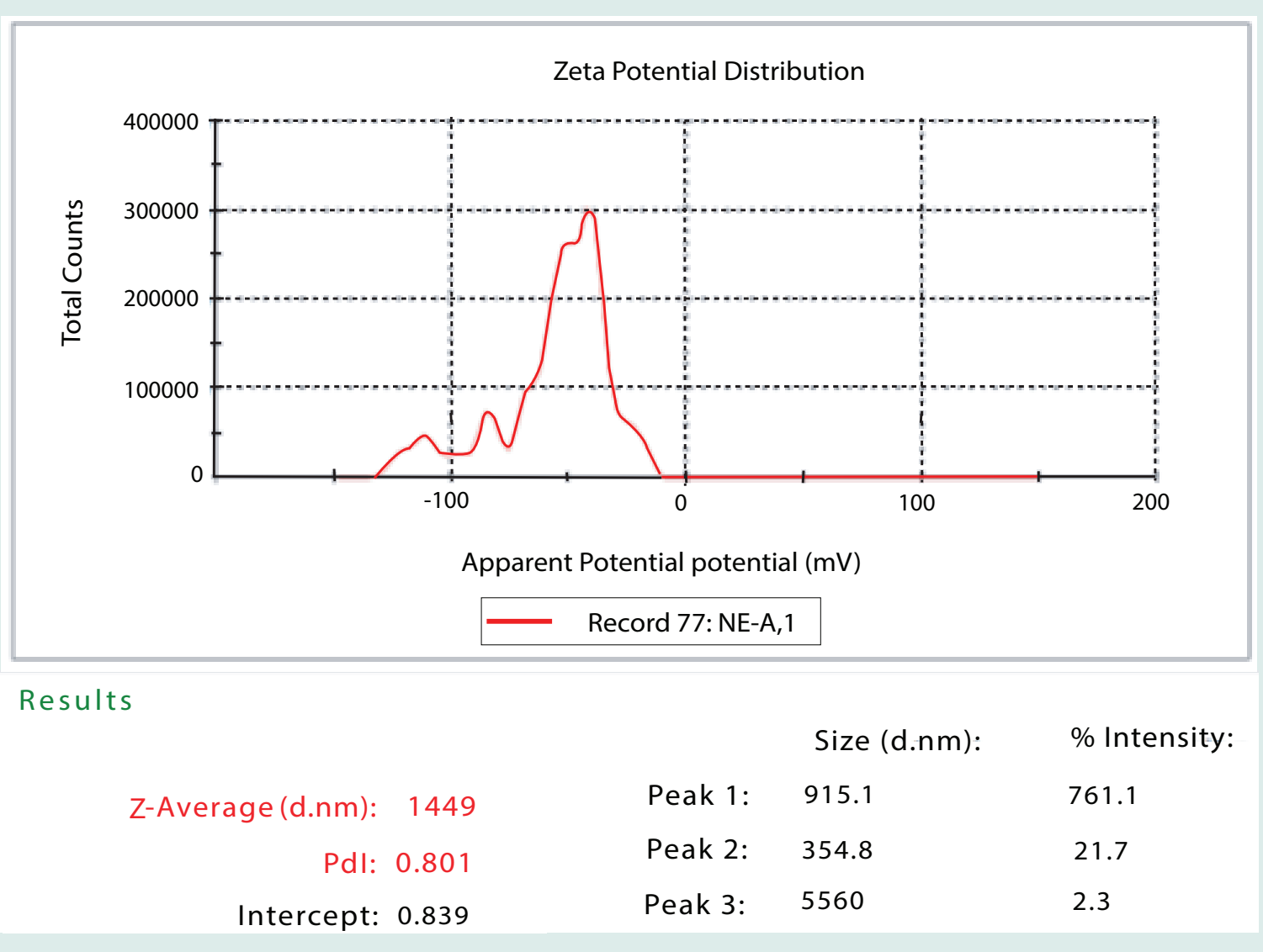
Nanoformulation

Test	Results
Visual inspection	Positive with stable emulsion
Conductivity	0.02 ± 0 µS
pH	7.19 ± 0.03
Viscosity	Relative viscosity = 2.778 ± 0.004 kg/m.s
Surface Tension	30.564 ± 0.002 Nm^{-1}

GC-MS analysis



Zeta potential

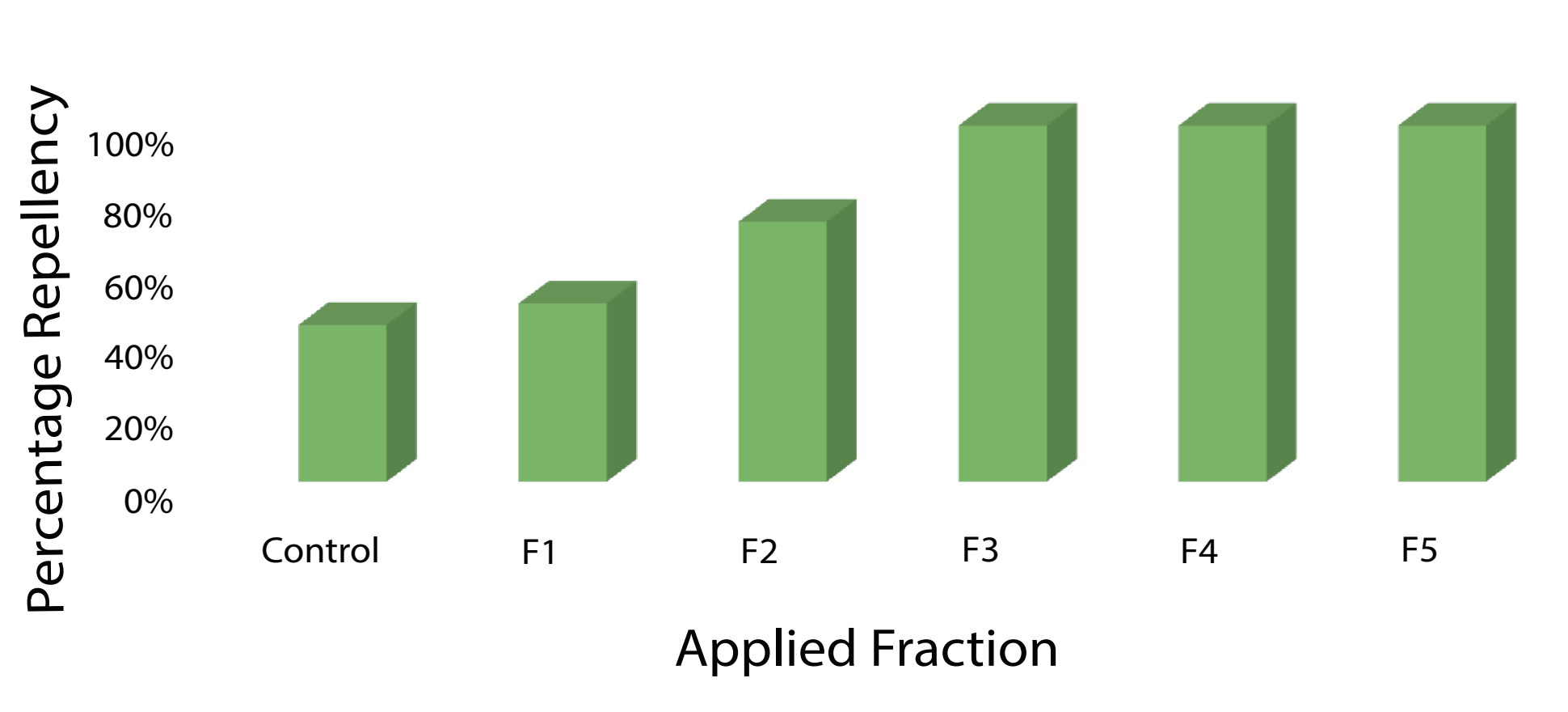


Results

Z-Average (d.nm): 1449
PdI: 0.801
Intercept: 0.839

	Size (d.nm):	% Intensity:
Peak 1:	915.1	761.1
Peak 2:	354.8	21.7
Peak 3:	5560	2.3

Mosquitoes Repellent activity



CONCLUSIONS

It can be concluded from the present results that oil in water emulsion system can be used as a medium to prepare *Ocimum tenuiflorum* essential oil loaded nanoemulsions with enhanced bioactivity. Also the formulated nanoemulsion can be a green alternative of the synthetic mosquito repellent.

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